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Research Article

Examining the Role of Prompt Engineering in Utilizing Generative AI Tools for Lesson Planning: Insights From Teachers' Experiences and Perceptions

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The purpose of this study is to explore the role of prompt engineering in utilizing GenAI tools like EduAid, Magic School, and ChatGPT to enhance lesson planning efficiency and content relevance. Additionally, it highlights the need for AI literacy and continuous professional development to align AI outputs with diverse student needs. A qualitative case study was used to collect data from 17 high school teachers from private and governmental schools. Document analysis, written reflections, and focus group talks were used as data sources to get detailed information about how the teachers used and thought about GenAI tools. The results show that effective prompts helped teachers use GenAI tools better, which makes lesson planning and material more relevant. Teachers emphasized that AI-generated material needed to be constantly improved to better meet the needs of all of their students. This study highlights the importance of developing AI literacy through continuous professional training to effectively enable teachers to integrate AI tools into their teaching practices.

Keywords: AI literacy; generative AI (GenAI) tools; prompt engineering; teacher perceptions

1. Introduction

Education is one of the many industries dramatically impacted by the quick development of technology [1]. Among these technological advancements, prompt engineering in GenAI tools has emerged as a significant innovation [2, 3]. GenAI tools, including EduAid, Magic School, and ChatGPT, employ artificial intelligence (AI) to produce content derived from user input [1, 4]. The effectiveness of these tools heavily depends on the prompts entered by users, highlighting the importance of prompt engineering [5]. Prompt engineering involves crafting precise and effective input to elicit AI systems' most valuable and relevant output [6]. This study examines the utilization of prompt engineering by teachers in Generative AI tools for lesson planning, as well as their experiences and perceptions regarding this practice.

Prompt engineering enhances the quality and relevance of AI-generated content, establishing it as a crucial skill for educators [7]. Mastering prompt engineering enables educators to effectively utilize AI tools, thereby optimizing time management, generating varied educational content, and customizing lesson plans to address the unique requirements of their students [8]. For example, teachers used different student-centered strategies to plan their lessons, such as the 5Es strategy, problem-based learning, project-based learning, or engineering design process [9]. Among these, the 5E model, commonly used by teachers, is widely recognized for its structured approach to lesson planning, focusing on the engage, explore, explain, elaborate, and evaluate stages [10, 11]. It guides students from the curiosity of asking questions and being engaged in their learning process to reflection and evaluation.

Research indicates the considerable potential of AI tools in education, especially in the automation and enhancement of lesson planning [9–12]. The effectiveness of these tools is significantly influenced by the quality of the prompts provided by users [13]. Prompt engineering is an essential competency that allows educators to derive the most pertinent and beneficial content from AI systems [2, 7, 13]. Although the potential is acknowledged, research specifically examining the role of prompt engineering in educational contexts remains limited. This study investigates the practical applications of prompt engineering within educational contexts, contributing to the development of essential AI literacy skills for the future. Educators must develop AI literacy skills to effectively integrate AI tools into their teaching practices [1, 14]. AI literacy encompasses comprehension of AI system functionality, effective interaction with these systems, and critical evaluation of AI-generated outputs [15]. Prior research highlights the significance of AI literacy in equipping educators and students for a technology-driven future [11, 14–16].

Regardless of the growing interest in AI in education, significant gaps remain in the literature regarding the practical applications of prompt engineering with GenAI tools for lesson planning [1]. For example, some studies have specifically examined how teachers interact with GenAI tools, the effectiveness of the prompts generated, and the overall impact on lesson-planning processes [2, 13, 17]. This study addresses these gaps by providing empirical evidence on using generative AI tools in educational settings and offering recommendations for their effective integration.

The current literature offers a broad view of AI's potential in education but often lacks specific insights into the practical challenges and strategies for effective prompt engineering. For instance, studies by Heston [2] and Nazary et al. [13] emphasize the importance of prompt quality but do not explore teachers' experiences crafting these prompts. Similarly, research by Chiu et al. [7] highlights the advantages of AI-generated content, yet it does not adequately address the iterative process educators engage in to refine prompts. This study seeks to address existing gaps by thoroughly examining teachers' experiences with GenAI tools, emphasizing the iterative process of prompt engineering and its influence on lesson planning.

Mastering prompt engineering is essential for AI literacy, allowing teachers to utilize GenAI tools in lesson planning. This study enhances existing literature by examining the development and application of AI literacy skills among teachers via prompt engineering. This study is essential due to the increasing incorporation of AI technologies in education and the imperative to assist teachers in enhancing their AI and digital literacy skills [18]. In the United Arab Emirates (UAE), there is a strong emphasis on incorporating digital tools to enhance educational outcomes, which aligns with the national agenda of UAE Vision 2021 [19]. The national agenda is aimed at establishing a first-rate education system using advanced technologies to improve teaching and learning outcomes. Furthermore, the study supports Sustainable Development Goal 4: Quality Education, which emphasizes the importance of inclusive and equitable education for all [20, 21].

Reflective practice is a critical component of teacher professional development. Schön's concepts of reflection-in-action and reflection-on-action provide a robust foundation for understanding teacher-reflective practices. Reflection-in-action refers to the process by which educators assess their experiences and implement immediate modifications to enhance teaching methods and results [22]. This prompt reflection enables educators to adjust their methods in real-time, thereby improving the learning experience as it progresses. Conversely, reflection-on-action necessitates that educators analyze their experiences postevent, facilitating comprehensive analysis and fostering long-term enhancements. This type of reflection facilitates a more thorough evaluation of teaching practices and outcomes, promoting continuous professional development.

Complementing Schön's theories, Kolb's experiential learning theory emphasizes learning through experience and reflection. According to Kolb [23], effective learning occurs through a cyclical process involving concrete experience, reflective observation, abstract conceptualization, and active experimentation. This cyclical process provides a structured framework for capturing teachers' reflective practices as they interact with AI tools for lesson planning. Kolb's model provides a comprehensive understanding of how teachers learn from their experiences and apply new knowledge to their practice.

Kolb's experiential learning model further delineates the process of experiential learning by incorporating stages of reflection and conceptualization [23]. This study builds on this literature by examining how teachers reflect on their use of AI tools both during the lesson planning process (reflection-in-action) and after completing their lesson plans (reflection-on-action). By integrating these theories, the study is aimed at providing a deeper understanding of teachers' experiences and insights, thereby informing the development of practical AI tools for educational purposes.

This study is aimed at investigating the effectiveness of teachers in employing prompt engineering when utilizing GenAI tools for lesson planning. This study evaluates the effectiveness of prompts created and utilized by teachers, their experiences with prompt engineering, and their perceptions of various GenAI tools in lesson planning. The following research questions guide the study:

1. How effective are the prompts entered by teachers in generating lesson plans using generative AI tools? How effective are the resulting outputs in meeting lesson planning objectives?

2. What are the teachers' experiences with using prompt engineering in generative AI tools to plan their lessons?

3. What are the teachers' perceptions of using prompt engineering in planning their lessons with different generative AI tools?

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2. Methodology

This study employed a qualitative case study design utilizing triangulation to explore the effectiveness of prompt engineering in GenAI tools for lesson planning. The qualitative approach is well-suited to gain in-depth insights into teachers' experiences and perceptions, particularly in educational research where context and participant perspectives are crucial [24].

This design is appropriate when the boundaries between the phenomenon and the context are unclear [25]. This study examines the effectiveness of prompt engineering within the context of teachers' plans utilizing GenAI. The case study method facilitates an in-depth analysis of this interaction in its natural context. This study utilized methodological triangulation to improve the validity and reliability of the findings, incorporating multiple data sources to examine the phenomenon [26]. Triangulation facilitates the cross-verification of data, yielding a more comprehensive and balanced understanding of the research problem [27].

2.1. Participants. The participants in this study were 17 high school teachers purposefully selected from private ($n1 = 7$, 41.18%) and governmental ($n2 = 10$, 58.82%) schools. The primary criterion for selection was the teachers' willingness to engage in the study and complete the tasks as outlined. This approach ensured that the participants had a genuine interest in exploring the potential of AI in their teaching practice.

The sample consisted of a greater proportion of female teachers, with 12 (70.59%) females and 5 (29.41%) males. The teachers instructed a variety of grade levels. Specifically, one (5.88%) taught Grades 8–9, one (5.88%) taught Grades 8–11, two (11.76%) taught Grades 8–10, and one (5.88%) taught both Grades 8 and 10. Other teachers indicated sole grade level teaching: 3 (17.65%) taught Grade 8, 2 (11.76%) each for Grade 11 and Grade 12, 2 (11.76%) taught Grade 10, and 1 teacher (5.88%) taught Grades 11–12. This indicates that the surveyed teachers cover a wide range of grade levels.

Science emerged as the most commonly taught subject, with nine educators (52.94%) instructing in biology, chemistry, physics, environmental science, and general science. Mathematics accounted for five teachers, representing 29.41% of the total. Each subject, including English, Computer Science, and Geography, was represented by a single teacher, accounting for 5.88%. This demonstrates a robust representation of STEM disciplines. Most teachers, 10 (58.82%), indicated 11–15 years of teaching experience. Of the participants, 7 (41.18%) possessed 6–10 years of teaching experience, suggesting that all individuals were relatively experienced in the field of education. The ages of the teachers exhibited considerable variation, categorized as follows: 3 (17.65%) aged 30–35 years, 7 (41.18%) aged 36–40 years, 6 (35.29%) aged 40–45 years, and 1 (5.88%) aged 46–50 years.

2.2. Instruments

2.2.1. Teachers' Task (Document Analysis). The task assigned to the teachers aimed to examine the utilization of GenAI tools by educators in lesson planning, thereby addressing the initial research question of the study. The assignment required developing a 20–30 min lesson plan utilizing backward design principles, focusing on defining end goals first and then planning in reverse to align objectives, activities, and assessments [28]. The aim is to document the prompts submitted by educators to AI tools and the corresponding outputs produced by these tools.

Teachers were instructed to create a lesson plan utilizing an AI tool with which they are familiar, including options such as EduAid.AI, Magic School, and ChatGPT. The document analysis primarily concentrated on the prompts submitted by the teachers and the resultant outputs produced by the AI tools in the sections outlined in Appendix A. In this task, teachers anticipated that AI tools would produce detailed, sequential lesson plans. However, several AI-generated outputs were overly broad, requiring additional modifications to fulfill specific classroom requirements.

2.2.2. Teachers' Written Reflection. The teachers' written reflections were an integral part of the study, providing valuable insights into their experiences with GenAI tools for lesson planning, which addresses the second research question of the study. Using Kolb's [23] experiential learning model, the teachers reflected on their activities, observations, challenges, and strategies and considered how their practices might evolve (see Appendix B for the reflective questions).

2.2.3. Focus Group Discussion. The focus group discussion is aimed at addressing the third research question of the study. The objective is to gain a deeper understanding of teachers' experiences and to collect qualitative insights that cannot be obtained solely through document analysis and written reflections (reflection-on-action).

The focus groups were conducted online using the Zoom conference platform, lasting 45–50 min. Ten teachers participated in the discussion, distributed equally into two groups, structured around semistructured, open-ended, reflective questions adapted from Kolb's experiential learning model. The questions were sent to two educational experts to determine their face validity and clarity. The experts compared the questions to the purpose of the study and provided feedback. They commended the questions for their alignment with the study's objectives and their ability to elicit detailed and meaningful responses from the participants (see Appendix C).

2.3. Procedure. A qualitative case study approach was employed utilizing methodological triangulation, incorporating data collection through document analysis, written reflections, and focus group discussions.

First, an initial meeting was conducted with the participating teachers to ensure clarity and consistency in task completion. The meeting outlined the study's purpose and presented detailed guidelines. Educators received an overview of the task specifications and the application of backward design in lesson planning. The study discussed the selected GenAI tools and the documentation procedures for input and output. After the meeting, teachers received

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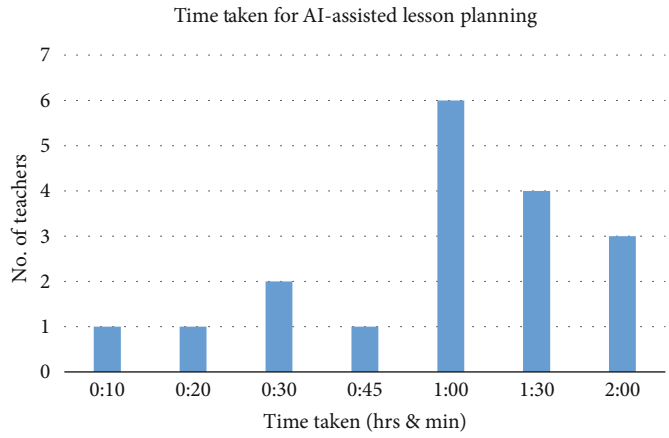


Figure 1: The distribution of time taken for AI-assisted lesson planning.

an online link where they could copy the prompts entered into the AI tools and a copy of the outputs generated by the AI tools.

Then, the participants completed a reflective questionnaire based on Kolb's experiential learning model [23], detailing their experiences and insights from using the GenAI tools (reflection-in-action). These reflections included descriptions of their activities, observations of the AI tools' effectiveness, conceptualizations for improvement, and plans for future use of AI tools in lesson planning. The reflections helped identify challenges faced in writing prompts for GenAI tools and how teachers modified their prompts to get better outcomes. The reflections were submitted through an online platform.

The focus group discussion was conducted online via the Zoom conference platform to examine teachers' perceptions regarding using GenAI tools (reflection-on-action). The data obtained from document analysis, written reflections, and focus group discussions were analyzed thematically to identify principal themes and subthemes [29]. This method facilitated the incorporation of various viewpoints and a thorough assessment of the efficacy of prompt engineering in GenAI tools. Integrating data from various sources facilitated a comprehensive and rigorous analysis, mitigating bias associated with relying on a single data type and aiding in validating the findings.

2.3.1. Ethical Considerations. The research ethical committee of Zayed University approved the study and assigned a Research Incentive Grant (RIF) no. 23071, with Ethics Application Number ZU23_068_F. Participants received consent forms at the beginning of the study. They were informed that their participation was voluntary, anonymous, and confidential and that they could withdraw from the study at anytime. Accordingly, informed consent was obtained from all participants, ensuring their voluntary participation and confidentiality.

3. Results

3.1. Analysis of Teachers' Tasks. Figure 1 illustrates the variability in the duration required to complete the AI-assisted lesson planning task. One hour was the duration selected by six teachers, representing 35.29% of the total, thus making it the most prevalent choice. Four teachers accounted for 23.53% of the sample, taking 1.5 h, while three teachers represented 17.65%, taking 2 h. Two teachers (11.76%) completed the task in 30 min, while one teacher each (5.88%) took 10, 20, and 45 min. This variation suggests differing levels of familiarity with GenAI tools.

Figure 2 shows that the teachers' experience with AI tools also varied, with six teachers (35.29%) rating their experience as 3 out of 5, another six teachers (35.29%) rating it as 4 out of 5, and 3 teachers (17.65%) rating it as 5 out of 5. Two teachers, representing 11.76%, assessed their experience with a 2 out of 5 rating. The distribution indicates that most teachers perceived themselves as having moderate to high experience with AI tools. The following is a sample of prompts submitted and produced in the tasks assigned to educators. Appendix D contains a comprehensive list of prompts.

3.1.1. Objectives. The Objectives section lists the goals that educators hope to accomplish with their lesson plans. The objectives produced by AI tools following multiple prompts from educators promote clarity and coherence in instructional goals. For example,

Prompt entered by T1: "Rewrite exponential functions to identify rates. Interpret the parameters of an exponential function within the context of compound interest problems. Construct exponential models given two points or by using regression."

Objective generated: "Students will be able to rewrite exponential functions to identify rates, interpret the parameters of an exponential function within the context of compound interest problems, and construct exponential models given two points or by using regression."

T3 entered: "Students will be able to demonstrate an understanding of osmosis in plant and animal cells, including its effects on cell structure and function."

Objective generated: "Students will understand the process of osmosis in plant and animal cells and how it impacts cell structure and function."

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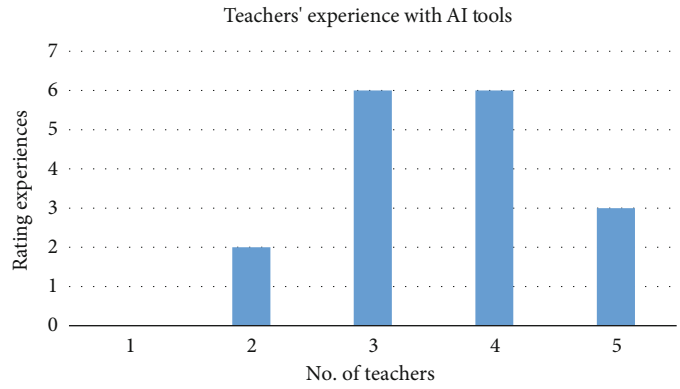


Figure 2: Teachers' experience with AI tools.

Similar prompts across different subjects highlighted the adaptability of the tools.

T9 entered: "Create objectives for a lesson on the principles of chemical bonding."

Objective generated: "Students will explain the types of chemical bonds and their properties."

3.1.2. Introduction. The introduction sets the stage for the lesson, capturing students' attention and providing context. The generated introductions were contextually relevant, often enhancing the teachers' initial prompts. For example,

T2 entered: "Begin by showing students a simple electrolytic cell setup. Ask students to brainstorm any applications they know of that use electrolytic cells."

Introduction generated: "Students will explore various applications that use electrolytic cells and understand the underlying principles of electrolysis."

T3 entered: "Show students the pictures of the potato and strawberry soaked in sugar solution. Ask students to make observations about the changes they see."

Introduction generated: "Students will engage in a hands-on exploration using a picture of a strawberry soaked in a sugar solution to observe and analyze the changes that occur over a period of time."

Other introductions included the following:

T9 entered: "How can I introduce the concept of chemical bonding to high school students?"

Introduction generated: "Start with a short video explaining the different types of chemical bonds followed by a class discussion."

3.1.3. Main Activity. The main activity is the core part of the lesson, where students engage in hands-on learning. The generated activities were interactive and diverse, ensuring student engagement and practical understanding. Examples include the following:

T1 entered: "Explore (8–10 minutes): Provide students with simple exponential functions and compound interest problems to solve in pairs."

Activity generated: "Think-Pair-Share Activity: Exponential Functions and Compound Interest. Step 1: Think. Step 2: Pair. Step 3: Share."

Similar prompts resulted in similarly effective responses.

T8 entered: "Design a hands-on activity for teaching Newton's laws of motion."

Activity generated: "Students will perform experiments to demonstrate each of Newton's laws using a cart and track."

T16 entered: "Develop a hands-on experiment for a lesson on acid-base reactions."

Activity generated: "Students will perform titrations to determine the concentration of an unknown acid or base."

3.1.4. Technology Integration and Differentiation. This section involves incorporating technology and addressing diverse learning needs. The generated suggestions were practical and tailored to enhance learning experiences. For example,

T2 entered: "Engage: Begin the lesson by showing students a real-life example of an electrolytic cell in action, such as the electrolysis of water to produce hydrogen and oxygen."

Output: "Integration of Technology: Utilize simulation software or virtual labs to allow students to explore electrolytic cells in a digital format."

T3 entered: "Provide a virtual lab platform where students can perform experiments."

Output: "Add links to high-quality animations and diagrams on osmosis."

T11 entered: "Develop a plan for using technology in a lesson on climate change impacts."

Output: "Use climate modeling software to predict future climate change impacts and scenarios."

3.1.5. Formative Assessment Planning. Formative assessments help gauge student understanding throughout the lesson. The generated assessments were diverse and aimed at continuous evaluation. Examples include the following:

T1 entered: "Evaluate (5–7 min): Provide students with a set of problems that require them to rewrite exponential functions, interpret parameters, and construct exponential models."

Output generated: "Students will be able to rewrite exponential functions to identify rates, interpret the parameters of an exponential function within the context of compound interest problems, and construct exponential models given two points or by using regression."

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TABLE 1: Quantitative summary of teachers' sentiments in teachers' written reflection.

Category	Positive	Negative	Neutral	Summary
Concrete experience (feel)	7 (41%)	1 (6%)	9 (53%)	Teachers found the tools *easy to use* and efficient, especially in *streamlining initial planning* (T1, T4). However, some teachers needed *more specificity* for their classroom needs (T6).
Reflective observation (watch)	6 (35%)	2 (12%)	9 (53%)	Teachers appreciated the *structure and efficiency* of AI tools but needed more *customization* (T9). Some noted the *generalized outputs* that did not align perfectly with their specific needs (T10).
Challenges/limitations	3 (18%)	10 (59%)	4 (23%)	Common issues were *lack of specificity* and the need for *multiple iterations* (T4, T9). Teachers struggled with adapting the tools to *diverse student needs* and *classroom contexts* (T12).
Prompts	10 (59%)	3 (18%)	4 (23%)	Teachers generally found *prompts useful* but often had to refine them for *specific needs* (T5, T9). Some felt that the tool needed more *refinement* and *specificity* to generate targeted content (T12).
Abstract conceptualization (think)	7 (41%)	2 (12%)	8 (47%)	Teachers praised the *structured approach* but needed *subject-specific adjustments* (T4). Some wanted the AI to provide more *interactive tools* and *customization* to meet *educational standards* (T9).
Active experimentation (do)	8 (47%)	1 (6%)	8 (47%)	Teachers planned to *refine prompts* and explore additional features for lesson planning (T6, T21). Some expressed interest in exploring *new AI tools* for specific educational contexts (T19).

Similar prompts were addressed with tailored formative assessments.

T8 entered: "Generate formative assessment ideas for a physics lesson on Newton's laws of motion."

Output generated: "Use quick quizzes and concept maps to assess understanding of Newton's laws."

T16 entered: "How can I assess students' understanding in a chemistry lesson on acid-base reactions?"

Output generated: "Utilize lab reports and interactive quizzes to assess understanding of acid-base reactions."

3.2. Teachers' Written Reflections. Teachers' evaluations of GenAI tools for lesson planning indicate a beneficial effect on efficiency and organization. Challenges concerning specificity, alignment with educational standards, and the necessity for refinement were identified. The reflections suggest significant potential for AI tools to improve educational practices, contingent upon their ongoing development to address the specific needs of educators and diverse student populations. Table 1 provides a quantitative summary of the teachers' sentiments, offering a comprehensive overview of the results, and the detailed responses from teachers are quoted.

3.2.1. Concrete Experience (Feel). Teachers engaged with various AI tools to create lesson plans, integrating these tools into their instructional strategies. The consensus among teachers was that GenAI tools were user-friendly and efficient in streamlining the initial stages of lesson planning.

Teachers used EduAid.AI for a 5E lesson plan and reported the following:

T1: "The preparation time was shorter and easy to find differentiated activities and hands-on activities."

T2: "The EduAid.AI helped in generating detailed lesson components. The tool was effective in providing comprehensive outputs. Prompt generation was efficient and easy."

Another teacher who used Magic School AI for a lesson on electrolysis stated the following:

T4: "The AI tool saved time, introduced ideas especially for assessment and differentiation. It turned the lesson to be easier and time saver."

Similarly, a teacher using ChatGPT noted the following:

T6: "ChatGPT helped me plan the entire lesson, including the activities like the Interactive Equation Relay Race, integrating technology such as interactive whiteboards and online tools."

3.2.2. Reflective Observation (Watch). Teachers' reflective observations indicated that AI tools positively influenced the lesson planning process by providing structure and efficiency. However, some teachers noted the necessity for customization to align with specific classroom needs.

Teachers using ChatGPT observed the following:

T6: "Using ChatGPT streamlined my lesson planning, providing quick and relevant prompts. It worked well to generate ideas and save time, though sometimes minor adjustments were needed."

T9: "Using ChatGPT to plan the lesson on solving equations with variables was beneficial overall. It efficiently generated ideas and structured the lesson around educational objectives like the Interactive Equation Relay Race and

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technology integration. However, I needed to refine and adapt its suggestions to fit specific classroom needs and ensure alignment with educational standards. ChatGPT's functionality was impressive for generating coherent prompts and offering insights into teaching strategies and technological tools. It streamlined the planning process and contributed to a well-rounded lesson plan that was engaging and educational for students."

Some teachers highlighted the content curation capabilities of EduAid.AI, stating the following:

T2: "The GenAI tool's functionality in content curation was invaluable, offering insights and perspectives that enriched our classroom activities."

Despite these benefits, some limitations were noted. One teacher mentioned,

T4: "The AI tool sometimes provided overly complex outputs,"

while others pointed out,

T10: "The AI tools offered generalized suggestions and prompts, which did not adequately address the diverse learning needs and abilities present in my specific classrooms."

3.2.3. Challenges/Limitations. Challenges and limitations were common themes in the reflections. Teachers faced issues with the specificity and relevance of AI-generated content, alignment with curriculum standards, and the need for manual adjustments.

Some teachers using Magic School AI reported the following:

T4: "The tool sometimes required multiple iterations to get the desired prompts."

Other teachers using ChatGPT reported the following:

T9: "Minor limitations in the AI's ability to fully understand context-specific needs."

Others who used EduAid.AI stated the following:

T2: "A few challenges in ensuring the prompts were suitable for all student levels."

Another teacher emphasized the general nature of AI outputs, stating the following:

T12: "…there were notable limitations and challenges. These included the need for personalization, addressing advanced and remedial needs, overcoming technical issues, and adding creativity to the lesson. Balancing AI efficiency with human insight and ensuring inclusivity and cultural relevance were crucial in creating an effective and engaging lesson plan."

These challenges highlight the need for more detailed and tailored AI-generated content to meet the specific needs of educators.

3.2.4. Prompts. The effectiveness of AI-generated prompts was generally positive, although multiple iterations were often required to refine the outputs. Teachers appreciated the guidance and structure provided by the AI tools.

A teacher using ChatGPT remarked the following:

T7: "The prompts generated by ChatGPT were effective in guiding the lesson planning process. They provided clear directions and examples that aligned with the educational goals."

T9: "My strategy was to write specific yet flexible prompts to foster creativity and adaptability in the classroom. ChatGPT generally understood the prompts well, though I refined them through multiple iterations to ensure they precisely met the lesson objectives and student needs. Overall, ChatGPT's prompts were instrumental in structuring the lesson and generating innovative ideas for activities, technology integration, and differentiation strategies."

Similarly, a teacher using Magic School AI noted the following:

T5: "The tool was able to understand the prompt and respond by the first, maximum the second prompt. I just [made] a few changes in each prompt to be able to find the details I'm looking for."

Teachers who used EduAid.AI mentioned the following:

T2: "I did not have a strategy to write the prompt but I kept adding more details in my prompt as if I am chatting with the website."

T11: "I carefully crafted prompts to ensure the generated content was relevant, comprehensive, and aligned with the 5Es strategy. By using clear, specific, and iterative prompts, I was able to refine the AI-generated content to meet the specific needs of my classroom……..Despite some initial limitations, the iterative approach allowed for customization and improvement, making the final lesson plan both effective and engaging."

These reflections suggest that while AI tools can generate helpful prompts, further refinement and specificity are often needed.

3.2.5. Abstract Conceptualization (Think). Teachers reflected on the overall improvement in their lesson planning process due to using AI tools. They acknowledged the efficiency and structured approach provided by these tools but also identified areas for further enhancement.

Some teachers using EduAid.AI noted the following:

T2: "The AI tool significantly improved the lesson planning process by providing a detailed, structured, and efficient framework aligned with the 5Es strategy."

T11: "I think that my lesson planning process is faster now and I have [a] clear vision of the lesson flow. I needed to use other options in this AI tool like generating differentiated worksheets for the elaborate part."

Others who used Magic School mentioned the following:

T4: "The tool was very helpful in planning and organization, giving ideas and elaboration. The tool will be very helpful if they could improve and make it more specified to the subject itself."

Some who used ChatGPT stated the following:

T9: "To further enhance its support for lesson planning, the AI tool could offer more customization options tailored to different educational standards and student needs, provide interactive feedback on initial prompts, ensure alignment with curriculum standards, and integrate visual or interactive tools for better lesson visualization and implementation."

Another teacher suggested the following:

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T12: "To further support lesson planning, the AI tool could enhance its ability to suggest differentiated prompts for diverse student needs and provide more detailed feedback on lesson structure and flow."

These insights highlight the potential for AI tools to evolve further and better support teachers in creating customized and effective lesson plans.

3.2.6. Active Experimentation (Do). Based on their experiences, teachers planned to refine their approach to using AI tools and explore additional features to enhance their lesson planning and teaching strategies.

Some teachers mentioned that

T6: "In the future, I'll focus on refining prompts early to better meet educational goals and student needs. I'll also explore AI tools that offer interactive feedback and enhance visual learning experiences."

T19: "I just need to know what I need to write and be specific to get clear results. I would like to try Magic School as I saw many educators recommending It in social media. I might also want to try a tool that generates [PowerPoint slides] for the lessons."

T21: "I think that using AI tools in the future will be very beneficial specially in interactive lab activities (as chemistry and science teachers); young students are not allowed to do all kinds of experiments in the lab for safety precautions, using simulations to the lab will make it easier for young students to perform all experiments virtually even the adults will help them to try different experiments that will be unsafe for them to do or try in the lab."

These reflections suggest that teachers should take a proactive approach to continually improving their use of AI tools, exploring new functionalities, and integrating advanced technologies to enhance their instructional practices.

3.3. Focus Group Analysis. The focus group discussions provided valuable insights into teachers' experiences with AI tools for lesson planning. Key emerging themes include concrete experience, reflective observation, challenges/limitations, prompts, abstract conceptualization, active experimentation, overall impact, and future use. These themes reflect educators' perspectives as they navigated the integration of AI in their teaching practice. Direct quotations from participants are used to illustrate these key points.

Overall, the analysis reveals that while AI tools offer significant benefits regarding idea generation and time efficiency, they also present challenges that require careful management. The results highlight the necessity for continuous improvement of AI technologies and focused teacher training to enhance their effectiveness in developing practical and innovative lesson plans. Table 2 provides a quantitative summary of the teachers' sentiments, offering a comprehensive overview of the results, and the detailed responses from teachers are quoted.

3.3.1. Tool(s) Used. Educators chose AI tools according to their particular requirements, with preferences differing based on the features provided by each tool. Magic School was selected due to its academic rigor and visual content, whereas EduAid was favored for its differentiation capabilities.

- T1 mentioned, "I used Magic School because it has the 5Es lesson plans in an academic way. It's very specific even though not all options are there."
- T2 added, "EduAid was suitable because it gave me exactly what I needed with special adjustments from my side."
- T3 noted, "I used both EduAid and Magic School. Magic School is great for visuals and normal worksheets, while EduAid is better for differentiation and is easier for teachers with more options."
- T4 shared, "I used ChatGPT for prompts and lesson plan transcripts, and Prezi because it's visually attractive for students."

3.3.2. Concrete Experience (Feel). Teachers shared their initial thoughts and feelings when introduced to using AI tools for lesson planning. They appreciated the time-saving aspects but noted that AI tools often required supplementary effort to meet specific teaching needs.

- T1 felt that "Using EduAid and Magic School made my life as a teacher easier. I have the general outline and know what to enhance. It also saves time for specific video questions."
- T3 highlighted a different aspect, saying, "I found AI tools limited and had to add my own questions to focus on specific lessons. Google Sheets were better because they could be edited and practiced with."
- T4 noted, "I'm not using AI mainly because I need specific pictures, but I use scenario-based questions. The AI doesn't differentiate between grade levels."
- T11: "I was familiar with Magic School AI from previous use, which motivated me to choose it for its detailed 5Es lesson plans."

3.3.3. Reflective Observation (Watch). Reflecting on their experiences, teachers noted the AI tools' varying effectiveness in enhancing lesson planning. While AI tools provided diverse suggestions, their utility often depended on the teacher's input and additional resources.

- T2 observed, "The AI tools gave me different types of questions based on students' levels, which saved time, but the game-based learning suggestions were often ineffective. Teachers-pay-teachers resources were more effective."
- T4 noted, "Using the AI tool helped with elaboration and generating deeper questions and skills, but it needed more specificity from my end to be truly useful."

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TABLE 2: Quantitative summary of teachers' sentiments in focus group discussions.

Category	Positive	Neutral	Negative	Summary
Tool(s) used	4 (50%)	4 (50%)	0 (0%)	Teachers selected tools based on specific needs: *Magic School* was valued for its *academic rigor* and *visuals* (T1), while *EduAid* was appreciated for *differentiation capabilities* (T2). Some used both for *different purposes* (T3), and *ChatGPT* was favored for *prompt generation* (T4).
Concrete experience (feel)	4 (50%)	2 (25%)	2 (25%)	Teachers found AI tools *time-saving* but noted that additional effort was required to meet *specific needs* (T1, T4). Some mentioned that the tools *lacked customization* for certain grade levels or subject-specific content (T3, T4).
Reflective observation (watch)	4 (50%)	2 (25%)	2 (25%)	AI tools provided *structure* and *efficiency*, but teachers needed to *adapt outputs* to fit classroom needs (T2, T9). Some found AI suggestions for *game-based learning* ineffective (T2).
Challenges/limitations	2 (25%)	1 (12.5%)	5 (62.5%)	Teachers reported *limited outputs* and the need for *multiple iterations* (T1, T3). AI tools struggled with *grade-specific prompts*, leading to inefficiencies (T1, T3, T11).
Prompts	5 (62.5%)	2 (25%)	1 (12.5%)	Teachers noted that *specific prompts* improved *accuracy* but required *multiple refinements* (T4, T12). Using *role-based prompts* (e.g., "imagine you are an engineer") led to better results (T4, T2).
Abstract conceptualization (think)	3 (37.5%)	5 (62.5%)	0 (0%)	Teachers reflected on AI tools as a *starting point* but emphasized the need for *traditional resources* and *expertise* (T2, T3, T11). AI provided *quick suggestions* but required additional materials to be effective.
Active experimentation (do)	5 (62.5%)	2 (25%)	1 (12.5%)	Teachers planned to be *more precise* with prompts and use *additional tools* to complement AI (T2, T1). Some expressed interest in exploring AI for *interactive feedback* and *learning tools* like *Kahoot* and *Quizzes* (T9).
Overall impact and future use	5 (62.5%)	1 (12.5%)	2 (25%)	Teachers recognized AI tools as *helpful* but stressed the need for *training* and *awareness* of their *limitations* (T1, T3). Future use will focus on *efficient tools* that integrate multiple lesson-planning resources (T10).

- T9 stated, "ChatGPT provided good outputs but they were not always accurate. It required significant refinement."

3.3.4. Challenges/Limitations. Participants encountered several challenges, particularly regarding the specificity and accuracy of the outputs.

- T3 explained, "The AI tools do not always give specific answers. I had to provide prompt after prompt and be very specific about the grade level. It also limits creativity."
- T1 shared a similar experience, "The outputs were limited and required me to be very specific. I had to keep trying different prompts, which was time-consuming. AI tools also limited creativity in lesson planning."
- T11: "Limited outputs required multiple iterations and often did not align perfectly with my lesson objectives."

3.3.5. Prompts. The effectiveness of prompts was a significant factor in the successful use of AI tools. Detailed and scenario-based prompts significantly improved the relevance and accuracy of AI outputs.

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- T4 elaborated, "Giving a role to the AI, like 'imagine you are an engineer,' resulted in better, scenario-based outputs."
- T2 added, "Using specific scenarios in prompts made the AI's suggestions more useful, but it still required multiple iterations to get satisfactory results."
- T12 stated, "The prompts needed to be very detailed to get accurate outputs. I found that giving AI a specific role or scenario improved the results."

3.3.6. Abstract Conceptualization (Think). Teachers reflected on how their lesson planning process improved with AI tools and suggested improvements. While AI tools provided a useful starting point, teachers emphasized the need to combine AI outputs with traditional resources and their expertise.

- T2 reflected, "AI tools improved my lesson planning in terms of lexile levels and activity suggestions. It's a rapid resource that can be used in limited time."
- T3 stated, "AI tools give a good starting point, but I still need other resources and my own expertise to create effective lesson plans."
- T11 stated, "The AI tool provided a good starting point and helped organize my thoughts, but I still needed to supplement it with additional resources."

3.3.7. Active Experimentation (Do). Teachers discussed how they plan to refine their use of AI tools in future lesson planning. Findings revealed that future use of AI tools will involve more precise prompts and a broader exploration of different tools to complement traditional methods.

- T2 mentioned, "In the future, I will be more precise and specific in prompts to get everything I need related to my students."
- T1 added, "I won't rely solely on AI tools. I will look for different ones, including those for editing videos and interactive tools like Kahoot and Quizzes."
- T9 stated, "I need not to rely solely on AI tools and will look for different ones that offer more specific functionalities."

3.3.8. Overall Impact and Future Use. Teachers reflected on the overall impact of AI on their lesson planning and their intentions for future use. While recognizing the benefits of AI tools, teachers stressed the importance of not relying on them entirely and called for comprehensive training for new educators.

- T1 stated, "AI tools are helpful to a degree but should not be relied upon entirely. New teachers need to be aware of their limitations and biases."
- T3 mentioned, "New teachers need guidance and training to use AI tools effectively. Experience helps in knowing what to adapt and use from AI outputs."

TABLE 3: Summary of the common themes identified from teachers' reflections and focus group discussions.

Theme	Positive	Neutral	Negative
Time efficiency	12 (71%)	3 (17%)	2 (12%)
Customization and flexibility	8 (47%)	4 (24%)	5 (29%)
User friendliness	10 (59%)	5 (29%)	2 (12%)
Content relevance	9 (53%)	4 (24%)	4 (24%)
Adaptability to classroom needs	7 (41%)	4 (24%)	6 (35%)
Prompt generation	11 (65%)	3 (18%)	3 (18%)
AI tool limitations	5 (29%)	3 (18%)	9 (53%)
Lesson plan structure	10 (59%)	5 (29%)	2 (12%)
Overall satisfaction	8 (47%)	5 (29%)	4 (24%)

- T10 stated, "I will explore AI tools that consolidate lesson planning resources into one cohesive platform for more efficient use."

3.4. Summary of the Results. Based on teachers' written reflections and the focus group discussions, some themes were identified and quantified in Table 3. The results presented indicate that teachers perceived AI tools as time-efficient, especially in the creation of lesson plan components. However, several teachers observed that the tools necessitated modifications and enhancements to fulfill particular instructional requirements. Customization emerged as a prevalent theme; whereas many valued the adaptability of AI tools, some encountered difficulties in modifying the outputs to correspond with their classroom environments and specific instructional objectives.

Iterative prompts were highlighted as a crucial element in enhancing the quality of AI-generated outputs. Teachers observed that more specific prompts were crucial for generating accurate and relevant outcomes; yet, several expressed irritation due to the continuation to enhance these prompts.

Teachers commended the user-friendliness of AI tools, which enabled fast adjustment (e.g., EduAid). However, several educators encountered some challenges in navigating the interface at first. As shown in Table 4, several educators considered the AI-generated content beneficial, while others emphasized that it frequently required additional modifications (e.g., grade level and subject-specific criteria) to align with their teaching standards and student needs.

4. Discussion

4.1. RQ1: How Effective Are the Prompts Entered by Teachers in Generating Lesson Plans Using Generative AI Tools? How Effective Are the Resulting Outputs in Meeting Lesson Planning Objectives? The analysis of teachers' tasks revealed notable variations in the duration taken to complete the AI-assisted lesson planning, with the most common duration being 1 h. This variation likely reflects differing levels of familiarity and proficiency with GenAI tools, a factor noted in previous research by Nazary et al. [13].

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TABLE 4: Summary comparison between AI tools used by teachers.

AI tool	Strengths	Limitations	Best use case
Magic School	- Academic rigor, structured plans (5E model) - Visual content for interactive lessons	- Lack of flexibility - Generalized content, requiring adjustments	Ideal for teachers seeking structured, academic-focused lesson plans with visual content
EduAid.AI	- Differentiation for diverse student needs - Detailed, customizable lesson plans	- Steep learning curve - Needs multiple iterations for desired results	Best for teachers needing differentiated lessons and customization for varying student needs
ChatGPT	- Fast idea generation - Flexible prompts, ease of integration with tech tools	- Requires refinement for specificity - Accuracy concerns	Ideal for time-saving tasks and flexible content generation, especially for tech integration

The variation in time taken reflects the iterative nature of prompt engineering, as described by Chiu et al. [7]. Teachers who required more time likely engaged in additional iterations to enhance their prompts, demonstrating the significance of iterative processes in obtaining high-quality AI-generated results. This finding highlights the necessity for thorough training in prompt engineering to improve the utilization of GenAI tools in lesson planning.

Most teachers assessed their experience with AI tools as moderate to high, indicating a general comfort with the technology. The presence of certain teachers needing additional support highlights the necessity for continuous professional development in AI literacy. This is consistent with existing literature [1, 14], which supports the need for educators to acquire skills essential for the effective integration of AI into teaching practices.

The evaluation of specific prompts and generated outputs revealed a high degree of coherence and alignment, especially in formulating instructional objectives. For example, when a teacher provided a prompt regarding exponential functions and compound interest, the AI's output aligned closely with the specified educational objectives. This indicates that well-designed prompts can effectively direct AI tools to generate pertinent and precise content, supporting the conclusions of Heston [2]. The adaptability of GenAI tools across various subjects was also apparent. Prompts in physics, chemistry, biology, and mathematics produced precise and contextually relevant learning objectives. A prompt for a high school physics lesson on Newton's laws produced an objective that explicitly outlined the application of these laws. The ability to generate subject-specific content aligns with the findings of Druga et al. [11], who highlighted the extensive applicability of AI in creating educational materials.

Moreover, the AI-generated introductions and main activities were particularly effective in enhancing lesson engagement and ensuring practical understanding. For instance, an engaging introduction for a lesson on Newton's laws of motion using a cart and track demonstration was directly aligned with the teacher's prompt. Other examples, such as hands-on activities for teaching osmosis and titrations in chemistry, demonstrated the AI's capability to generate varied and practical learning tasks. These findings suggest that AI tools can significantly enhance lesson introductions, making them more engaging and contextually relevant, which is critical for effective teaching [7, 15].

The AI-generated suggestions for technology integration and differentiation were practical and specifically designed to improve learning experiences. Incorporating virtual lab simulations for a lesson on cellular respiration and offering interactive online tutorials for a mixed-ability class on chemical bonding were effective strategies that catered to diverse learning needs. The findings are consistent with the work of Steinbauer et al. [16], which examined the role of AI in facilitating personalized learning environments.

Planning for formative assessment was enhanced by AI-generated outputs, which provided a range of continuous evaluation methods. Exit tickets and peer evaluations for assessing understanding of the osmosis supplied educators with practical tools to measure student comprehension during the lesson. This aligns with the literature highlighting the role of AI in enhancing formative assessment practices [10].

4.2. RQ2: What Are the Teachers' Experiences With Using Prompt Engineering in Generative AI Tools to Plan Their Lessons?

4.2.1. Concrete Experience (Feel). Teachers reported a positive impact of GenAI tools such as EduAid.AI, Magic School AI, and ChatGPT on lesson planning, indicating that these tools enhanced efficiency in the initial stages of the process. This is consistent with the research conducted by Long and Magerko [14] and Varghese and Chapiro [1], which emphasizes the capacity of AI tools to improve educational practices by promoting efficiency and organization. EduAid.AI demonstrated effectiveness in generating detailed lesson components, facilitating teachers' access to differentiated activities, thereby supporting Varghese & Chapiro [1]. Magic School AI was recognized for its efficiency, particularly in generating ideas for assessment and differentiation. The findings indicate that GenAI tools can substantially decrease the time and effort needed for lesson planning, enabling teachers to concentrate more on instructional strategies and student engagement, as noted by the Author (2023).

Although the tools were user-friendly and beneficial, additional refinement was frequently required to ensure outputs matched particular classroom requirements. This

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underscores the iterative aspect of prompt engineering, as described by Chiu et al. [7], highlighting the necessity of refining prompts to attain high-quality AI-generated outputs.

4.2.2. Reflective Observation (Watch). Observations revealed that AI tools enhanced lesson planning through improved structure and efficiency. Teachers utilizing ChatGPT observed that it facilitated the prompt generation process efficiently, although slight modifications were occasionally required to ensure alignment with educational standards and particular classroom settings. This observation corroborates Heston's [2] focus on the importance of prompt quality and the iterative process required for achieving optimal outputs.

Several educators emphasized the effectiveness of EduAid.AI in content curation, recognizing its significant role in enhancing classroom activities. Challenges arose when AI-generated content required manual adjustments to align with specific classroom contexts, highlighting the significance of relevance and alignment with educational objectives, as noted by Nazary et al. [13].

Despite these advantages, certain limitations were observed. Educators noted that AI tools occasionally generated excessively complex outputs or generalized recommendations that failed to sufficiently meet the varied learning needs and abilities within particular classrooms. This underscores the need for more specific and customized AI-generated content to address the particular requirements of educators, aligning with the conclusions of Druga et al. [11] regarding the continuous evolution of AI tools to accommodate varied educational demands.

4.2.3. Challenges/Limitations. Teachers encountered several challenges with AI tools, particularly regarding the specificity and relevance of AI-generated content. Issues such as needing multiple iterations to refine prompts and difficulties aligning AI outputs with curriculum standards were common. Another challenge is the AI's ability to fully understand context-specific needs. This finding aligns with the literature on AI literacy, emphasizing the need for educators to develop robust AI literacy skills to effectively integrate AI into their teaching practices [1, 14].

4.2.4. Prompts. The effectiveness of AI-generated prompts was mostly favorable; however, multiple iterations were frequently necessary for refinement. Teachers valued the direction and framework offered by the AI tools, which played a crucial role in producing creative concepts for activities, technology integration, and differentiation strategies. This underscores the significance of high-quality, prompt engineering, as Heston [2] and Chiu et al. [7] highlighted. Furthermore, the refinement of prompts to obtain precise and relevant outputs is emphasized as essential, corroborating the findings of Nazary et al. [13] regarding the significance of prompt quality in the efficacy of AI tools.

4.2.5. Abstract Conceptualization (Think). Teachers acknowledged the efficiency and structured approach provided by AI tools but also identified areas for further enhancement. Those using EduAid noted improvements in lesson planning efficiency through a detailed framework aligned with the 5Es strategy, supporting literature on the benefits of AI-generated content [10, 11].

Others who used Magic School and ChatGPT suggested that the AI tools could be further improved by offering more customizable options tailored to different educational standards and student needs, reflecting Steinbauer et al. [16] on the need for personalized learning environments facilitated by AI tools.

4.2.6. Active Experimentation (Do). Teachers aimed to improve their utilization of AI tools by concentrating on initial prompt refinement and investigating supplementary features to augment lesson planning. There was an expression of interest in AI tools that provide interactive feedback and improve visual learning experiences. These reflections indicate that teachers should adopt a proactive approach in consistently enhancing their utilization of AI tools by investigating new functionalities to improve instructional practices. This is consistent with existing literature on AI literacy and emphasizes the necessity for continuous professional development to assist educators in the effective integration of AI technologies into their teaching practices [18, 20].

4.3. RQ3: What Are the Teachers' Perceptions of the Use of Prompt Engineering in Planning Their Lessons With Different Generative AI Tools?

4.3.1. Tool(s) Used. Teachers selected various AI tools based on their specific needs and preferences, demonstrating the flexibility and adaptability of these technologies in educational settings. Some teachers preferred Magic School's academic rigor and visual content, while others favored EduAid's differentiation capabilities. This highlights the role of AI tools in providing tailored educational content, aligning with Chiu et al. [7], who emphasize the benefits of AI-generated content in enhancing lesson plans. Additionally, some teachers combined tools to leverage different functionalities, supporting Steinbauer et al.'s [16] advocacy for a multitool approach to utilize AI technologies in education fully. This highlights the need for a multitool approach to fully use AI technologies in education, as suggested by Steinbauer et al. [16], who advocate for personalized learning environments facilitated by AI tools.

4.3.2. Concrete Experience (Feel). Teachers observed that AI tools facilitated lesson planning by offering organized and effective frameworks. This aligns with the findings of Long and Magerko [14], which emphasize the efficiency improvements resulting from integrating AI tools in educational practices. Some educators observed that AI-generated content lacked specificity, indicating that supplementary resources such as Google Search were occasionally more effective for obtaining detailed information. This is consistent with the findings of Nazary et al. [13], which highlight the necessity for iterative refinement of AI-generated content to address diverse educational needs adequately.

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4.3.3. Reflective Observation (Watch). Teachers noted that although AI tools provided many suggestions, their efficacy was inconsistent. AI tools generated various questions tailored to student levels; however, specific suggestions necessitated additional refinement for enhanced utility. This highlights the necessity of tailoring AI-generated content to meet particular classroom requirements, as Druga et al. [11] noted. Furthermore, certain educators emphasized the dependability of conventional resources compared to AI tools, indicating that a hybrid approach is essential, as Varghese and Chapiro [1] corroborated.

4.3.4. Challenges/Limitations. Participants recognized various challenges, especially concerning the specificity and accuracy of the outputs. Teachers needed to offer various prompts and specify the grade level to attain the intended outcomes, illustrating the iterative aspect of prompt engineering as emphasized by Heston [2].

Furthermore, manual modifications were frequently required to adapt AI-generated content to particular classroom contexts, highlighting the constraints of existing AI tools in meeting varied educational demands and the essential function of teacher expertise, consistent with the findings of Chan & Hu [15].

4.3.5. Prompts. The effectiveness of prompts significantly impacted the successful use of AI tools. Teachers noted that targeted, scenario-based prompts yielded better results, supporting the conclusions of Nazary et al. [13], who emphasized the importance of prompt quality. Detailed and iterative prompts were often necessary to improve AI-generated outputs, demonstrating the ongoing refinement process described by Chiu et al. [7].

4.3.6. Abstract Conceptualization (Think). Teachers recognized advancements in their lesson planning methodologies attributed to the utilization of AI tools, while also pinpointing areas necessitating further development. Although AI tools have enhanced efficiency, Varghese and Chapiro [1] noted that teachers highlighted the necessity for supplementary resources and expertise to develop effective lesson plans. This emphasizes the complementary function of AI tools alongside the essential role of teacher expertise in education, consistent with the findings of Varghese and Chapiro [1].

4.3.7. Active Experimentation (Do). Teachers reflected on potential modifications to their utilization of AI tools moving forward, demonstrating a proactive stance towards enhancing and tailoring their application of AI in educational contexts, as indicated by Steinbauer et al. [16].

Several teachers indicated a desire to investigate AI tools with specialized functionalities, highlighting the necessity of continuously adapting AI technologies to address the changing demands of education. This aligns with the World Bank's [18] findings regarding the essential role of continuous professional development in AI literacy.

4.3.8. Overall Impact and Future Use. Teachers evaluated the influence of AI tools on their understanding of AI in educational contexts and their feasibility for consistent integration in lesson planning. The findings highlight the necessity of harmonizing AI integration with professional expertise to facilitate effective and innovative lesson planning, as supported by Varghese and Chapiro [1]. The necessity for guidance and training for new teachers was highlighted, emphasizing the importance of continuous professional development in facilitating the effective integration of AI technologies into teaching practices, as noted by the UAE National Committee on SDGs [20].

5. Implications

The implications of this study are that, even with rich sources of lesson planning in GenAI tools, effective use entails a combination of continuous professional development, AI literacy, and an iterative model for prompt engineering. The results confirm that effective engineering of prompts helps teachers make a more direct link between lesson content and educational objectives. Besides, AI tools can reduce cognitive overload during lesson planning, and teachers can then devote their full attention to instruction and student activity. Teachers learned that AI output, even when useful, sometimes required refinement and retuning in an attempt to perfectly tailor it to specific individual classroom requirements. This underlines not only utilizing AI tools but also developing competencies for effective communication with such tools through effective prompt engineering. Iterative refinement of prompts is an important part of getting high-quality, relevant output, according to the study.

From a pragmatic perspective, the study promotes additional training programs for teachers in AI literacy, particularly in terms of prompt engineering. Teachers with proper training in developing effective prompts experienced easier incorporation of AI tools in lesson planning. However, tools must become even more specific and tailor-made in an effort to cater to the variegated and ever-changing demands of a classroom environment. Teachers have expressed a desire for more flexible AI capabilities that can respond to changing educational standards, grade levels, and subject-specific requirements. Additionally, this work highlights the necessity for a balanced model in which AI tools work to amplify, not supplant, teacher expertise. AI can, for example, work to make parts of lesson planning easier, but teachers will remain at the center of curating information in a way that makes it relevant, correct, and sensitive to students' needs. As AI technology continues to mature, teachers' expertise in tailoring AI output for use in their individual educational environments will become ever more critical.

6. Limitations

This study has several identified limitations. The study included teachers who volunteered to participate, which may introduce self-selection bias. The presence of teachers with prior interest or proficiency in AI tools may bias the results, leading to more favorable experiences. The study

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examines particular generative AI tools, such as EduAid, Magic School, and ChatGPT, which may not encompass the functionalities of other existing tools. This limitation may restrict the insights obtained concerning AI's role in lesson planning. The study's emphasis on high school teachers and particular grade levels may restrict its relevance to educators in other educational contexts, including elementary or tertiary education. The sample size of 17 high school teachers restricts the generalizability of the results. Further study is required to use a larger and more heterogeneous sample to validate the findings across other educational contexts. The study depends on self-reported data from teachers, which may be influenced by personal biases or inaccuracies in memory regarding their experiences.

7. Recommendations

This study recommends educator training that emphasizes key areas to improve the effectiveness of generative AI tools in lesson planning. Comprehensive AI literacy programs must be implemented to ensure that teachers grasp the fundamentals of AI and demonstrate proficiency in prompt engineering, especially in the iterative refinement process. Training should emphasize the role of AI tools in facilitating personalized learning through customized strategies that address the varied needs of students. The training should assist teachers in tailoring AI outputs to fit particular classroom contexts, thereby ensuring that AI enhances their professional expertise. Educators should utilize AI tools to develop formative assessments, including peer evaluations and exit tickets, to facilitate ongoing student evaluation. The training should prioritize the role of AI in optimizing lesson planning, thereby minimizing time allocated to administrative duties, enabling educators to concentrate on instructional methods, and enhancing student engagement.

8. Conclusion

This study provides critical insights into the significant role of prompt engineering with GenAI tools in lesson planning, emphasizing its potential to improve the educational process substantially. The findings demonstrate that practical and iterative prompt engineering allows teachers to maximize the potential of GenAI tools, including EduAid, Magic School, and ChatGPT, to develop more impactful and pertinent lesson content. Effective prompt engineering facilitates the development of customized lesson plans that cater to varied student needs and enhance efficiency in the planning process, as evidenced by the research of Chiu et al. [7] and Nazary et al. [13].

The findings indicate that although GenAI tools offer teachers organized and efficient solutions, manual refinement of AI outputs is frequently required to align with particular classroom needs. This highlights the significance of AI literacy in education, especially the iterative refinement process essential for enhancing AI-generated content [2, 7]. Teachers reported favorable experiences with these tools, indicating that prompt engineering enhanced the quality and relevance of lesson outputs [1]. Nonetheless, specific challenges, including the AI's restricted capacity to adequately address the diverse abilities of students, were also observed [11].

This study emphasizes the necessity of ongoing professional development in AI literacy, corresponding with the essential requirement for educators to effectively incorporate AI tools into their instructional methods [1, 14]. The evolution of AI necessitates that educators maintain proficiency in developing prompts and refining AI outputs to align with educational standards. The World Bank [18] emphasized that fostering AI literacy skills is essential for educators to promote inclusive and equitable learning environments.

This study confirms that prompt engineering is essential for educators to optimize using GenAI tools. Although these tools provide significant advantages in time efficiency and content creation, the expertise of teachers is critical for refining AI outputs to align with curriculum standards and address specific student needs. As GenAI shapes the future of education, it is necessary to equip educators with prompt engineering skills and AI literacy to promote high-quality, personalized, and inclusive education [13, 21].

Appendix A: Teachers' Task

The teachers were asked to design a lesson plan using specific GenAI tools: EduAid.AI, Magic School, and ChatGPT. The primary focus of the document analysis was on the prompts entered by the teachers and the corresponding outputs generated by the AI tools in the sections mentioned as follows:

1. Objective(s): Teachers began by defining clear, specific learning objectives for their lesson. These objectives were entered into the AI tools to generate relevant content and strategies.
2. Introduction: Teachers described how they would introduce the topic to their students. They used the AI tools to generate engaging and contextually appropriate introductions.
3. Main activity: Teachers planned hands-on or explorative tasks for students. They utilized AI-generated suggestions to outline the main activities that would help achieve the learning objectives.
4. Technology integration and differentiation: Teachers integrated technology into their lesson plans and ensured that activities were adapted to meet diverse learning needs. The AI tools provided ideas for incorporating technology and differentiating instruction.
5. Formative assessment planning: Teachers designed formative assessments to gauge student understanding throughout the lesson. AI tools offered suggestions for question types and formats suitable for the learning objectives.

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Appendix B: Teachers' Written Reflection

The reflective questions were provided to the teachers.

1. Concrete experience (feel)
    - o. What did you do? Describe the activity, the AI tool used, and how it was integrated into the lesson.
    - o. What was the experience of using the AI tool for planning the lesson?
    - o. How easy was the tool to come up with the prompts?
2. Reflective observation (watch)
    - o. Reflect on how the AI tool influenced your planning process. What worked well and what did not?
    - o. Consider the functionality of the AI tool. What insights did it provide?
3. Challenges/limitations
    - o. Were there limitations/challenges to the planning process? Please elaborate.
4. *Prompts*
    - o. Elaborate on the prompts you came up with. Did they work? Why?
    - o. Did you have a strategy to write the prompts?
    - o. Did the tool understand the prompt or how many iterations of the prompts did you do?
5. Abstract conceptualization (think)
    - o. Based on your experience and observations, do you think your lesson planning process improved? Elaborate.
    - o. How can the AI tool used be further improved to support your lesson planning?
6. Active experimentation (do)
    - o. Considering what you learned from this planning process, how might you alter your approach to using AI tools in the future?
    - o. What other AI tools or features could you explore?

Appendix C: Focus Group Discussion Questions

The questions used in the focus group were as follows:

1. Tool(s) used
    - What tool(s) have you used in your lesson planning? Why these tools?
2. Concrete experience (feel)
    - Can you describe your initial thoughts and feelings when you were first introduced to the task of using an AI tool for lesson planning?
    - How accurate was the AI tool (in response to your prompts) in generating outputs and suggestions? Were there any difficulties in getting the desired outputs?
3. Reflective observation (watch)
    - Reflecting on the process, how did the AI tool influence your lesson planning? What aspects of the tool worked well, and what did not?
    - What insights did you gain about the functionality and usefulness of the AI tool during your planning?
4. Challenges/limitations
    - What were the main challenges or limitations you faced while using the AI tool for lesson planning? How did you address these challenges?
5. Prompts
    - How effective were the prompts you came up with? Did you have a specific strategy for crafting these prompts, and how many iterations did you need to get satisfactory results?
6. Abstract conceptualization (think)
    - Based on your experience, do you feel that your lesson planning process improved with the use of the AI tool? How so?
    - How could the AI tool be further improved to better support teachers in lesson planning?
7. Active experimentation (do)
    - Considering what you learned from this experience, how might you change your approach to using AI tools in the future?
8. Overall impact and future use
    - How has this experience influenced your perception of AI in education?
    - What advice can you give to new teachers?

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Appendix D: Analysis of Teachers' Tasks

Objectives

TABLE A1: The goals teachers aim to achieve through their lesson plans.

#	Prompts entered	Prompts generated
T1	"Rewrite exponential functions to identify rates. Interpret the parameters of an exponential function within the context of compound interest problems. Construct exponential models given two points or by using regression."	"Students will be able to rewrite exponential functions to identify rates, interpret the parameters of an exponential function within the context of compound interest problems, and construct exponential models given two points or by using regression."
T2	"Students will explore various applications that use electrolytic cells and understand the underlying principles of electrolysis."	"Prepare a 5E lesson plan that includes a lab activity related to electrolytic cells, ensuring all levels of students are engaged."
T3	"Students will be able to demonstrate an understanding of osmosis in plant and animal cells, including its effects on cell structure and function."	"Students will understand the process of osmosis in plant and animal cells and how it impacts cell structure and function."
T4	"Students will be able to solve linear equations with variables on both sides by applying inverse operations and properties of equality, and verify the solutions by substituting them back into the original equations."	"This objective aligns with the Common Core State Standards for Mathematics (CCSS.MATH.CONTENT.8.EE.C.7), which focuses on solving linear equations in one variable, including equations with coefficients represented by letters."
T5	"Reflect on and analyze today's argumentative article, identify key arguments, and engage in activities to explore the topic."	N/A
T6	"To teach the topic of wave interference to grade 8 students."	"Using NGSS Standards, prepare a lesson plan for grade 8 on the topic of wave interference."
T7	"Calculate the area of a parallelogram."	"Calculate the area of a parallelogram by multiplying the base by the height."
T8	"Generate learning objectives for a high school physics lesson on Newton's laws of motion."	Students will understand and be able to apply Newton's three laws of motion to solve problems."
T9	"Create objectives for a lesson on the principles of chemical bonding."	"Students will explain the types of chemical bonds and their properties."
T10	"What should be the objectives for a biology class on cellular respiration?"	"Students will describe the process of cellular respiration and its importance in living organisms."
T11	"Develop objectives for an environmental science lesson on climate change impacts."	"Students will analyze the impacts of climate change on different ecosystems."
T12	"List learning objectives for teaching quadratic functions in a high school algebra class."	"Students will solve quadratic equations and understand their applications."
T13	"Create objectives for a lesson on the fundamentals of computer programming."	"Students will write and debug basic programs using a programming language."
T14	"Generate goals for a high school lesson on statistical data analysis."	"Students will analyze and interpret statistical data using appropriate tools."
T15	"Develop objectives for a geography lesson on geographical information systems (GIS)."	"Students will use GIS tools to analyze geographical data."
T16	"What should be the learning objectives for a chemistry class on acid-base reactions?"	"Students will understand the principles of acid-base reactions and perform titrations."
T17	"Create objectives for a physics lesson on electromagnetism."	"Students will explain the principles of electromagnetism and solve related problems."

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Introduction

TABLE A2: The stage for the lesson, capturing students' attention and providing context.

#	Prompts entered	Prompts generated
T1	"Engage (5–7 min): Show a video clip of real-life examples of compound interest and exponential growth."	"Same as previous"
T2	"Begin by showing students a simple electrolytic cell setup. Ask students to brainstorm any applications they know of that use electrolytic cells."	"Students will explore various applications that use electrolytic cells and understand the underlying principles of electrolysis."
T3	"Show students the pictures of the potato and strawberry soaked in sugar solution. Ask students to make observations about the changes they see."	"Students will engage in a hands-on exploration using a picture of a strawberry soaked in a sugar solution to observe and analyze the changes that occur over a period of time."
T4	"Hook: 'Imagine you and your friend are playing a video game where you both start with different amounts of points, and each round you earn or lose points. By the end of the game, both of you have the same total points. How can you figure out how many points you each had at the beginning? Today, we are going to learn how to solve problems like this by solving equations with variables on both sides."	"What type of an introduction can be used? Keep in mind that the lesson will be taught to a boys' section."
T5	"Using an audio-visual I found on YouTube that matches my topic."	"I have provided the prompt and did not need the AI to enhance it."
T6	"to activate their prior knowledge and by discussion method"	"Using NGSS Standards, prepare a lesson plan for grade 8 on the topic of wave interference. 1. Introduction (5 minutes): Hook: Show a short video clip demonstrating wave interference patterns in water and light. Discussion: Briefly discuss observations and introduce the concept of wave interference."
T7	"Warm up activity to start the lesson."	"Engage students in an introductory activity that explores the concept of finding the area of a parallelogram using manipulatives and real-world examples."
T8	"Create an engaging introduction for a lesson on Newton's laws of motion."	"Begin with a demonstration using a cart and track to illustrate Newton's laws of motion."
T9	"How can I introduce the concept of chemical bonding to high school students?"	"Start with a short video explaining the different types of chemical bonds followed by a class discussion."
T10	"What is a good way to start a lesson on cellular respiration?"	"Introduce cellular respiration with a model of a cell and a discussion on energy production in cells."
T11	"Develop an introduction for a lesson on climate change impacts."	"Show a video on the effects of climate change and discuss observable impacts with students."
T12	"How can I introduce quadratic functions to high school students?"	"Begin with a real-world problem that can be solved using quadratic functions."
T13	"Create an introduction for a lesson on computer programming fundamentals."	"Start with an interactive coding exercise where students write their first simple program."
T14	"What is an engaging way to start a lesson on statistical data analysis?"	"Introduce statistical data analysis with a real dataset and discuss its significance."
T15	"How can I introduce the topic of geographical information systems to high school students?"	"Begin with a demonstration of GIS software showing local geographical data."
T16	"What is a good way to start a chemistry lesson on acid-base reactions?"	"Start with a simple experiment demonstrating an acid-base reaction."
T17	"Develop an introduction for a physics lesson on electromagnetism."	"Begin with a demonstration using magnets and coils to illustrate electromagnetism."

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Main Activity

TABLE A3: The main activity is the core part of the lesson where students engage in hands-on learning.

#	Prompts entered	Prompts generated
T1	"Explore (8–10 min): Provide students with simple exponential functions and compound interest problems to solve in pairs."	"Think-Pair-Share Activity: Exponential Functions and Compound Interest. Step 1: Think. Step 2: Pair. Step 3: Share."
T2	"Divide students into small groups. Provide each group with materials to set up their own electrolytic cell."	"How can I integrate technology? and ensure activities are adapted for different learning needs?"
T3	"Design an engaging and challenging activity that incorporates various levels of difficulty and interests for students to explore the osmosis process in both animal and plant cells."	"Comprehension: How does osmosis contribute to the overall health of plant and animal cells? Application: Given a scenario, demonstrate how osmosis plays a role in maintaining cell turgor pressure in plant cells."
T4	"Main Activity: Equation Relay Race. Objective: To solve linear equations with variables on both sides and check the solutions."	"Main Activity: Interactive Equation Relay Race. Objective: Students will solve linear equations with variables on both sides using technology and differentiated support."
T5	"Create a chart on the board to list the key arguments and their supporting evidence."	"N/A"
T6	"Creating wave interference patterns using slinkies."	"Hands-On Activity: Slinky Experiment: In small groups, students use a slinky or wave rope to create waves and observe interference."
T7	"Team-Based activity using geometric hands-on to explore how to evaluate the area of a parallelogram."	"Students will collaborate in teams to develop strategies for determining the area of a parallelogram through hands-on exploration and problem-solving activities."
T8	"Design a hands-on activity for teaching Newton's laws of motion."	"Students will perform experiments to demonstrate each of Newton's laws using a cart and track."
T9	"What is an interactive way to teach chemical bonding principles?"	"Students will build models of different types of chemical bonds using molecular kits."
T10	"Create a main activity for a lesson on cellular respiration."	"Students will conduct an experiment to measure the rate of cellular respiration in yeast."
T11	"Develop a group activity for a lesson on climate change impacts."	"Organize a project where students analyze data on climate change impacts and present their findings."
T12	"What hands-on activity can help students understand quadratic functions?"	"Students will use graphing calculators to plot quadratic functions and find their roots."
T13	"Design an activity for a lesson on computer programming fundamentals."	"Students will complete a series of coding challenges to practice basic programming concepts."
T14	"Create an interactive activity for a lesson on statistical data analysis."	"Students will work in groups to analyze a dataset and present their findings using statistical tools."
T15	"What is an engaging main activity for teaching GIS?"	"Students will use GIS software to analyze and interpret geographical data from their local area."
T16	"Develop a hands-on experiment for a lesson on acid-base reactions."	"Students will perform titrations to determine the concentration of an unknown acid or base."
T17	"Create a main activity for a physics lesson on electromagnetism."	"Students will build and test electromagnets to explore the principles of electromagnetism."

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Technology Integration and Differentiation

TABLE A4: The incorporation of technology and addressing diverse learning needs.

#	Prompts entered	Prompts generated
T1	"Elaborate (12–15 minutes): Provide students with more complex scenarios involving compound interest and exponential growth."	"Students will work in pairs to solve simple exponential functions and compound interest problems."
T2	"Engage: Begin the lesson by showing students a real-life example of an electrolytic cell in action, such as the electrolysis of water to produce hydrogen and oxygen."	"Integration of Technology: Utilize simulation software or virtual labs to allow students to explore electrolytic cells in a digital format."
T3	"Provide a virtual lab platform where students can perform experiments."	"Add links to high-quality animations and diagrams on osmosis."
T4	"How can I embed Technology Integration and Differentiation."	"The main activity will change accordingly. Main Activity: Interactive Equation Relay Race."
T5	"Students read the argumentative article on a digital platform."	"During the guided analysis, I use an interactive whiteboard to chart key arguments and evidence."
T6	"For this topic we can use simulation."	"Reduce this plan to a 50-minute class period and include differentiated activities to cater to diverse learning styles."
T7	"Use a video for visual learners to explore and then give differentiated questions relevant to the video for them to answer."	"Identify questions for the following video link to be used for visual learners."
T8	"Suggest technology tools to enhance a physics lesson on Newton's laws of motion."	"Use simulation software to model and visualize Newton's laws in different scenarios."
T9	"How can I differentiate instruction for a mixed-ability class on chemical bonding?"	"Provide interactive online tutorials and simulations to cater to different learning paces."
T10	"What technology can I use to support a lesson on cellular respiration?"	"Incorporate virtual lab simulations to explore cellular respiration processes."
T11	"Develop a plan for using technology in a lesson on climate change impacts."	"Use climate modeling software to predict future climate change impacts and scenarios."
T12	"How can I use tech tools to differentiate a math lesson on quadratic functions?"	"Utilize graphing software and apps to visualize and manipulate quadratic functions."
T13	"What are some digital resources for teaching computer programming?"	"Access online coding platforms and interactive tutorials for practice and assignments."
T14	"How can I integrate technology into a lesson on statistical data analysis?"	"Use statistical analysis software and online databases for data exploration."
T15	"What technology tools can enhance a lesson on GIS?"	"Incorporate GIS software and online data repositories for hands-on learning."
T16	"How can I use technology to support a lesson on acid-base reactions?"	"Use virtual lab simulations to practice acid-base titration experiments."
T17	"What technology can I use to enhance a physics lesson on electromagnetism?"	"Integrate online simulations and virtual labs to explore electromagnetism concepts."

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Formative Assessment Planning

TABLE A5: Formative assessments help gauge student understanding throughout the lesson.

#	Prompts entered	Prompts generated
T1	"Evaluate (5–7 min): Provide students with a set of problems that require them to rewrite exponential functions, interpret parameters, and construct exponential models."	"Students will be able to rewrite exponential functions to identify rates, interpret the parameters of an exponential function within the context of compound interest problems, and construct exponential models given two points or by using regression."
T2	"First assessment: Students will write a brief report detailing an electrolytic cell application of their choice."	"Second assessment: Predicting the products of electrolysis in different scenarios."
T3	"Design a differentiated formative assessment to evaluate students' understanding of osmosis in both plant and animal cells through hands-on experiments and written analysis."	"Exit tickets summarizing key points learned during the class. Peer evaluation of written analysis of osmosis experiments."
T4	"Formative Assessment Planning: Pre-Assessment: Objective: Gauge students' prior knowledge and understanding of solving equations with variables on both sides."	"Formative Assessment Planning: Interactive Equation Relay Race. Real-time Monitoring: Circulate among groups during the activity to observe students' problem-solving strategies."
T5	"N/A"	"I tailored the assessments specifically to meet the needs of my students by selecting question types and formats that align with our learning objectives."
T6	"DOK Level questions."	"Include Depth of Knowledge (DOK) level questions to assess students' understanding of the topic at different cognitive levels."
T7	"Create a formative assessment worksheet to involve different types of questions."	"An inclusive worksheet with multiple choice, complete, and open-ended questions with differentiated levels."
T8	"Generate formative assessment ideas for a physics lesson on Newton's laws of motion."	"Use quick quizzes and concept maps to assess understanding of Newton's laws."
T9	"What are some quick checks for understanding in a chemistry lesson on chemical bonding?"	"Implement exit tickets and class discussions to check for understanding of chemical bonding concepts."
T10	"Develop formative assessments for a biology lesson on cellular respiration."	"Use problem-solving exercises and peer assessments to gauge understanding of cellular respiration."
T11	"How can I assess students' understanding during a lesson on climate change impacts?"	"Have students complete reflection journals and participate in group discussions."
T12	"What are some formative assessment strategies for a math class on quadratic functions?"	"Conduct mini-assessments using whiteboards and interactive math software."
T13	"Create formative assessments for a lesson on computer programming fundamentals."	"Use code review sessions and debugging exercises to assess programming skills."
T14	"What are effective formative assessments for a lesson on statistical data analysis?"	"Incorporate data analysis projects and peer review sessions to assess statistical skills."
T15	"What formative assessments can I use for a lesson on GIS?"	"Use project-based assessments and GIS software tasks to check for understanding."
T16	"How can I assess students' understanding in a chemistry lesson on acid-base reactions?"	"Utilize lab reports and interactive quizzes to assess understanding of acid-base reactions."
T17	"What formative assessments are effective for a physics lesson on electromagnetism?"	"Implement performance-based assessments where students demonstrate concepts through experiments and projects."

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Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Conflicts of Interest

The authors declare no conflicts of interest.

Author Contributions

All the authors: conceptualization, methodology, software, formal analysis, investigation, data curation, writing—-original draft, writing—review and editing, visualization.

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